
MINCO 430/440

Intelligent controller Manual



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I. Summarization

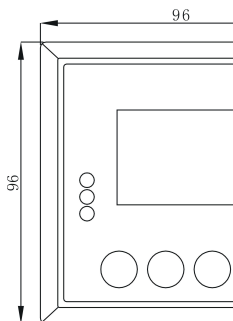
MINCO 430/440 controller adopts high performance microprocessor and industry components. It has measuring, controlling, protection, remote control, flexible software setting functions and user-defined to the input and output port, it can display all the measuring parameter, control parameters and the running state. The anti-jamming ability is strong, actually can meet different types of engine auto control requirements.

II. Characteristics

1. Minco430 apply for single generator automatic control system;
2. Based on 430, Minco440 adds mains monitoring, apply for double power automatic control system;
3. Double processing chip, virtual measurement of voltage and current, multifunction, smart action;
4. Wide-screen LCD display with 128*64 lattice and back-light;
5. Chinese and English double language menu, mutual operation, All the setting and operation can be completed without computer;
6. Auto start, auto protection, auto supply control;
7. Prefect auto protection, display warning message and work state, failure record more than 50 items;
8. Plenty of connected parameters (coolant temp, double oil pressure, oil temp. oil level etc.);
9. All relay contact capability is above 10A/250VAC/30VDC;
10. User-defined input/output, timer start & stop generator;
11. RS485 communication, attached "four remote control" monitor software;

III. Fixup dimension drawing

Operating Panel	W 96 x H 96mm
Installing Hole	W 85 x H 85mm
Depth	D 57mm



View Map

IV. Function definition and operation

4.1. Operate panel function instruction

Operate panel is composed of 128x64 LCD display, operation keys with state indicator light, and system menu operate press keys three parts.

(1) LCD display

Controller runs in normal (not setting state or not fault state), display all the measure parameters and present status of engine. Display screen can be alternately through pressing “set” button

Operation	Description
Main screen 1 (This page is just for 440)	Normal: 00.0HZ L-N: 000 000 000V
Main screen 2	F:00.0HZ PF:0.00 L-N:000 000 000V 00.0 00.0 00.0A Power 0000.0 KW
Main screen 3	Speed : 0000 RPM Battery: 25.8 V Energy: 00000.0 Run Time: 00000.00
Main screen 4	Coolant Temp.: --- °C Oil Pressure: --- MPa Fuel Level: --- % Charger: 00.0 V
Main screen 5	<u>STOP/OFF STATUS</u> <u>11-12-03/09:12:15</u>

Note:

① When “display change mode” set in “auto” switch state, LCD display screen will switch to next page for each 10 seconds; when “background light control” set in “auto” state, LCD screen background light will auto turn off without any operation after three minutes. Once the fault appear or press any key the background light turns on. LCD display may be failed during background light is off. Not seen as a fault. When “Background light” control setting as “constant light”, LCD background light will keep lighting.

② Not connecting with coolant temp. sensor, oil pressure sensor and fuel level sensor, it displays “—”, but that would have no influence on controller itself.

(2) Operation button

Content	Function
	Press “I” button, the above green LED will keep bright, the controller is in “start” state, start the generator in manual and keep running.
	Press “AUTO” button, the above yellow LED will keep bright, controller works in “auto” mode. ①When the “Remote start” switch is off and mains resume, the generator set will be stopped after cool down delay. ②When “Remote start” switch is closed or mains failure, the generator set will be delay started. ③ If timing start generator is valid, generator is be timing started. After the end of the timed start, will delay shutdown.
	Press “O” button, the above red LED will keep bright, controller works in “STOP / RESET” mode, it will activate “GENERATOR COOLDOWN”. De-Energize the clutch, slow to idle, activate Cool down timer and then stop the generator after the timer expires. During “GENERATOR COOLDOWN”, the indicator will flash and will keep bright after stop.

(3) System menu keys

Press “OK” button, holding 10 seconds, then to parameter setting menu, here the “O” button is redefined to exit “←”, “auto” key is redefined to add “↑”, “I” button redefined to reduce “↓”, “OK” key redefined to “→”

Content	Function
	Parameter setting /enter to next menu/confirm to revise
	Exit/back to the previous menu
	Page up the menu/ increase value
	Page down the menu/ decrease value

(4) State Indicator Light

Content	Function
	Indicate failure of generator, protected stop, fault content display in the LCD screen. When it's locked by the reason of start failure, over speed, low oil pressure or other faults, should press “O” button to restart the generator.
	Indicate warning information, which has no influence on generator normal working. Alarm detail will display on screen, such as low battery, etc. When multiple alarms, it will be displayed alternately on the LCD screen.

	Indicate “Remote Start” input status. It’s usually used for mains monitoring.
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4.2 Connection port definition

Port No	Function
Power Supply 8~36VDC, normal working current <300mA	
1	“-” Battery cathode input
2	“+” Battery anode input
Analog Input (input voltage range 0~5.0VDC)	
6	Coolant temp. sensor
7	Oil pressure sensor
8	Fuel level sensor
18	Excitation/charge failure input (charge engine D + port input, forbid to connect ground)
Speed Sensor Input	
19	Speed sensor input 1
20	Speed sensor input 2
Genset Three phase load current input (0~5A AC,with inside isolation)	
21	Load Current Phase A
22	Load Current Phase B
23	Load Current Phase C
24	Load Current Common ICOM
Genset Three Phase Voltage Input (0~300V AC, with inside isolation)	
9	Genset Neutral Line
10	Genset Voltage phase W
11	Genset Voltage phase V
12	Genset Voltage phase U
Mains three phase voltage input (0-300VAC, without insulation inside)	
This function is just for 440	
32	Mains voltage phase A
33	Mains voltage phase B
34	Mains voltage phase B
35	Mains zero line N
Switch Output Port (Relay insulated, start, fuel contact capability is 10A/250VAC/30VDC,others are 5A)	
13	Crank
14	Fuel

15	Common port (common port for fuel and CRANK contact)
3	Auxiliary Output 1
4	Auxiliary Output 2
5	Auxiliary Output 3
16、17	Auxiliary Output 4 (Passive Contact)
Switch Input Port (input port with optical coupler short circuit to GND is active)	
25	Remote Start
26	Auxiliary Input 1
27	Auxiliary Input 2
28	Auxiliary Input 3
29	Auxiliary Input 4
RS485 Communication Port	
30	RS485+
31	RS485—

V. Parameter set

All the parameters not only can be set on the spot by the controller operation panel key, but also can be read and written through communication port, details see communication protocol.

Press  key	Enter to Parameter Setting Interface		
	Switch inputs	Alarm limit set	Timing start set
	Relay outputs	Measure regulate	Coolant temp. set
	Shutdown Record	Delay time set	Oil pressure set
Press  or  key	Date and time	System set	Fuel level set
	Select examine/setting parameter content (reversed display when selected)		
Press  key	Enter to next menu of selected item		
Press  key	Exit the parameter setting		

Attention: If not pressing any keys within three minutes, it will automatically exit parameter setting, to avoid any illegitimate operation on the controller.

5.1. Parameter setting instruction

Inputs status	Real time display of input status	
	Remote run: 0	Emergency stop: 0
	Auxiliary input 1: 0	Auxiliary input 2: 0
	Auxiliary input 3: 0	Auxiliary input 4: 0
Attention: Press any key will exit		

Outputs status	Real time display of output status Crank: 0 Fuel: 0 Auxiliary output 1: 0 Auxiliary output 2: 0 Auxiliary output 3: 0 Auxiliary output 4: 0 Attention: Press any key will exit
Shutdown Record	Shutdown record 01/04 (Fault serial number/total number) Genset Crank failure!! (shutdown record) 09-06-03/11:26:38 (fault time) Attention: Press ↑, ↓key, display up and down fault record; press ←or→ will exit.
Date and time set	Date: year—month—day/week Time: hour-minute-second Press ↑, ↓key to change the reverse display data; Press ←reverse display move to the left, move to the first position then press ←then back to the superior menu, date and time will not changed; press →reverse display move to the right, move to the last position press →then back to the superior menu, date and time display are according to the new setting.
Alarm Limit Set	High Voltage: 0250 Low Voltage: 0200 Over Load_C:0450 High Frequency:056 Low Frequency:045 Over Speed: 1650 Over Power: 0500 High Coolant temp:0096 High fuel level:0080 Low Fuel Level: 0020 Low Battery:105 Low Charger:080 Low Oil pressure:020 Press ↑or↓, choose content and content reversed display; press ←back to the superior menu; Press →to enter choosing parameter setting, the selected parameter is underlined, it means the parameter is being operated. The first bit of this parameter reversed display, shows the data of bit can be changed. Enter the parameter setting state, press ↑, ↓key to change the reversed display data; press ←move reversed bit to the end of left, press ←and back to the superior menu , parameter will be not changed; press →reversed display move to the end of right , press →and back to the superior menu, parameter changed and saved. Attention : the unit of frequency is 0.01Hz, charge voltage and battery voltage is 0.01V, oil level is 0.01Mpa, power is 1KW and current is 1A.

Measurement -nt Calibration	Input password: 8421(default) Generator A:0000 Battery Voltage:0256 Generator B:0000 Charge Voltage:0000 Generator C:0000 Coolant Adjustment: 0060 Current A:0000 Oil pressure Adjustment: 0999 Current B:0000 Oil level Adjustment: 0100 Current C:0000 Mains phase A: 0000 Mains phase B: 0000 Mains phase C: 0000 Note: The measure regulate of mains three phase is just for 440 Password Authentication Input Method Press [+], [-] to change the reversed display data; press [←] to move the selected content to the left, press [→] key and back to the superior menu when the selected content move to the first bit, press [OK] key, move the selected to the end of right, enter the password press [OK], if the password is correct then get through the next menu. Users according the error value of the controller measuring data and the real data to decide whether you need to data adjust. The controller already adjusted before leaving factory, but it may be some warp in the using environment, if the warp is within the error range, we suggest not adjust the data, especially the genset current and power. Press [+], [-] choose content reversed display, press [←] back to superior menu; press [set] enter to choose data adjustment state, and the adjusting parameter underline. ,it means the parameter is being operated. The first bit of parameter reversed display, shows the data of bit can be changed. Input data can be with a tolerance of ± 10. Enter to data adjusting state, press [+], [-] to change the data, press [→] key, the reversed display turn left. when move to the first bit, press [←] then back to the superior menu, data adjustment is cancelled. Press [OK] the reversed display turn right, move to the fourth position press [OK] back to the superior menu, data adjustment achieved, parameter changed saved. For three phase voltage, three phase current and battery voltage adjustment, enter data adjust state, change the data
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	then press OK to finish the data adjustment.(Current keep two decimal fraction) . Minco 430/440 controller provides three phase voltage adjustment, three phase current adjustment, battery voltage adjustment and charge voltage adjustment. The measurement of controller will increase (reduce) according to the increased (decreased) value of adjustment. The scope of adjustment is $\pm 10\%$. But adjustment of coolant temp.,oil pressure and fuel level are different. Special explain, for coolant temp.,oil pressure, fuel level sensors maybe positive modulus(it means the sensor output add along with input added) , it maybe negative modulus(it means the sensor output minish along with input added) , add or minish adjust value lead to adjust result which is decided by the real situation Attention: the unit of current, battery voltage and charge voltage is 0.1(e.g. actual value=displayed vulae *0.1), oil pressure is 0.01MP.
Delay time set	Password input: 8421 (default) Cool down:020 Aux.input2 delay:005 Geset start:003 Aux. input3 delay:002 Crank interval delay:015 Aux. input 4 delay:005 Crank time:008 Loss speed delay:030 Bypass time:020 Shut time delay:000 Energize to stop:000 Transform delay:002 Pre-fuel delay:005 Over current delay:003 Idle start delay:010 Over voltage delay:003 Idle stop delay:015 Over frequency delay:003 Acceleration Time:000 Warming up delay:010 Aux. input 1 delay:003 Deceleration delay:020 Press ↑、↓ choose content reversed display, press ← back to superior menu; press → enter to choose data adjustment state, and the adjusting parameter underline. it means the parameter is being operated. The first bit of parameter reversed display, shows the data of this bit can be changed. Enter to data adjusting state, press ↑、↓ to change the data of reversed display; press ← to move the right, reach to the first bit then press → to back to the superior menu, data adjustment in valid; press → key, the reversed display turn

	right, when move to the end ,press  key to back to superior menu, the change of parameter is saved. Delay time up limit can't be over255 seconds , if setting is over 255 seconds system will change to 255 seconds automatically.																					
System Parameters Set	Input password: 8421 (default) <table><tr><td>Trip Speed:0350</td><td>Output 3: 002</td><td>Display Mode:0</td></tr><tr><td>Current ratio:0500</td><td>Output 4:008</td><td>Language C/E:0</td></tr><tr><td>Password:8421</td><td>Input 1:002</td><td>LCD mode :1</td></tr><tr><td>Address:120</td><td>Input 2:001</td><td>Speed source :0</td></tr><tr><td>Gear teeth:030</td><td>Input 3:006</td><td>Phase/Line :1</td></tr><tr><td>Output 1:000</td><td>Input 4:008</td><td>Mains failure start: 0</td></tr><tr><td>Output 2:006</td><td>Crank Mode: 0</td><td></td></tr></table> Note: The function(mains failure start) is just for 440. Press, choose content reversed display, pressback to superior menu; presskey, enter the setting state, the adjusting parameter is underline, it means the parameter is being operated. The first bit of parameter reversed display, show the data of this bit can be changed. After enter into the setting state, press, key to change the data, presskey to turn left, move to the first bit then pressto back to the superior menu, the parameter will not be changed; presskey to turn right, move to the end then presskey to back to the superior menu, parameter changes are saved.	Trip Speed:0350	Output 3: 002	Display Mode:0	Current ratio:0500	Output 4:008	Language C/E:0	Password:8421	Input 1:002	LCD mode :1	Address:120	Input 2:001	Speed source :0	Gear teeth:030	Input 3:006	Phase/Line :1	Output 1:000	Input 4:008	Mains failure start: 0	Output 2:006	Crank Mode: 0	
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Output 1:000	Input 4:008	Mains failure start: 0																				
Output 2:006	Crank Mode: 0																					
Timing Start Setting	Date: Month—Day/Week T-1: begin time: minute—end time: minute T-2: begin time: minute—end time: minute T-3: begin time: minute—end time: minute Press, key to change the data of reversed display; pressto turn left, when the reversed display move to the first bit of the left, presskey to back to the superior menu, the date and time will not be changed; pressto turn right, when the reversed display move the end of the right, presskey to back to the superior menu, the date and time setting will be saved.																					

Cooltant temp. Set/ Oil pressure Set/ Fuel level set	Input password: 8421 (default) 1_ 0.66V / 120 °C 5_ 1.94V / 70 °C 2_ 1.04V / 100 °C 6_ 2.36V / 60 °C 3_ 1.27V / 090 °C 7_ 3.00V / 40 °C 4_ 1.62V / 080 °C 8_ 4.09V / 10 °C Setting of coolant temp. oil pressure, fuel level sensor are similar. Every sensor has eight data point, sixteen data, which was displayed in two pages. Press  key to change the data of reversed display; press  to turn left, when the reversed display move to the first bit of the left, press  key to back to the superior menu, the data will not be changed; press  to turn right, when the reversed display move the end of the right, press  key to back to the superior menu, the data setting will be saved. Different sensor has different curves, MINCO430/440 controller can deposit curve data from different sensors to achieve common to different sensor. (eg. coolant sensor, takes eight typical temperature and corresponding internal resistance values R, MINCO430/440 controller with 150 ohm pull-up resistor, based on partial pressure of formula $5R/(150+R)$, calculates the voltage values corresponding to different temperature point, input the controller (must be voltage from low to high order). As per the measured voltage values and interpolated eight voltage/temperature point, controller calculates the actual temperature value. (below the minimum temperature or higher than the maximum temperature points Controller displays minimum temperature or maximum temperature data.)

5.2. System Parameter Description

Trip Speed	If examines the generator frequency>trip frequency when starts generator, it will consider the generator starts successfully and stops the crank output(trip speed generally is set to 1 / 3 of generator rated frequency).
Current Ratio	Current ratio setting value is corresponding to 5 , for example the current ratio setting in 500 , it's correspond with 500: 5.
Change Password	Leave factory password 8421 , please change the password on your own.
Address	Only use for multiple equipment networking control , in order to differentiate the equipment.
Gear tooth	Unused

Pre-fuel: Output will be energized during the period of pre fuel;

Warm-up: Output will be energized after genset at rated speed and before genset load run;

Acceleration/Deceleration: Output will be energized during the acceleration delay and deceleration delay, which coordinate to finish the mechanical speed governing;

Over Speed: Output will be energized when the genset is over speed;

Over Load: Output will be energized when genset is overloaded or overcurrent.

Rated Speed: Output will be energized when genset running at rated speed;

Battery: Output will be energized when low voltage;

Pumping: Output will be energized when low fuel level is measured to alarm, output will be vanished when high fuel level is measured, fuel can be resupplied automatically;

Alarm: Output will be energized when genset alarm;

Over heat: Output will be energized when temp. exceed coolant temp. upper.
Output will be vanish when temp. is 20 °C lower than the water temperature upper limit.

Mains: Control mains port of ATS.

Auxiliary Input Definition Instruction:

Monitor: Not control, only monitor the state, needless input ports can be set up to monitor;

Low Oil Pressure: Genset will shutdown when low oil pressure is measured;

High Genset Temperature: Genset will shutdown when the high genset temperature is measured;

Acceleration Limit/Deceleration Limit: Coordinate with output of acceleration and deceleration to finish mechanical speed governing

High Oil Temperature: Genset will protect to stop when high oil temperature is detected;

Low Fuel Level: Alarm when low fuel level is detected, but genset will not stop;

High Fuel Level: Realize pumping function with low fuel level together;

Float Failure Alarm: Float failure alarms during the running time when this port is closed, but genset will not shutdown.

Alarm: User-defined alarm, instruct to auxiliary input 1—4 alarm;

Alarm Non-stop: User-defined alarm, but only during the period of genset working, instruct to auxiliary input 1—4 alarm;

Alarm Stop: User-defined alarm, cause to shutdown when genset is running, instruct to auxiliary input 1—4 stop;

Remote Reset: Reset fault when genset stop normally if it has been protected



to stop, it's equivalent to press "O" button again.

Attention:

Display, alarm and protection of coolant temp.,oil pressure and fuel level can be realized by measuring the analog volume , and also can be realized by defining the high coolant temp.,low oil pressure, fuel level to the auxiliary input portion. If the analog volume and alarm input are exist together in the system, then any of alarm can come into being protection alarm. If the switch alarm protection is no need, please define the auxiliary input to another function; if the analog volume alarm protection is no need, please set the alarm up-low limit to the measurement limit so as to not alarm.

Output of acceleration and deceleration are in coordination with limit of acceleration and deceleration to realize mechanical speed adjustment. Normal speed adjustment function can be realized by definite output of acceleration and deceleration but not detect properly and not alarm.

Since output and input can be user-defined, Minco 430 controller's input and output function actualized is much more than the real amount of input and output. Although some function of controller has been appointed, which can't execute wherever there is no definition to input and output corresponding. For example, when genset normally running, if the remote start input closed, the genset will work on load. But if no supply output definition, there is no supply action, it would mean the genset has being run normally able to load..

5.3Delay Time Instruction

Delay of "cool down"	When the controller is in "Auto"state, once the "Retransformation Delay" finish or auto start finish, then to delay, genset will stop after delay.
Delay of "genset start"	When the controller is in "Auto" state, once the "Remote start" switch off ,then to delay, genset will stop after delay.
Delay of "Crank Interval"	When the cranking time delay finish, if the start succeed condition is not satisfied and can not reach the crank times limit, the delay will be repeated and crank times add 1.
Delay of "start running"	When the genset starts and begins to delay, if the start succeed condition is satisfied(genset frequency>trip frequency), it's consider to be genset start successfully and stop delaying.
Delay of "bypass time"	After the genset start successfully, begin to start the delay of bypass , "low oil pressure","high generator

	temp.”etc. will not be monitored during the delay to avoid any mistaken alarm when genset in initially starting.
Delay of “energize to stop”	Output of auxiliary stop relay is energized when genset stop.”Ergize to stop” delay begin,output of auxiliary stop relay is de-energized when delay finish.
Delay of “pre-fuel”	Delay of pre-fuel has begun before genset starts and relay of pre-fuel close at the same time., Relay of pre-fuel cutoff after delay finish,and genset start to crank.
Delay of “idle start”	Delay of idle start begin after the genset starts successfully,and the relay of “idle start” begin to work at same time.
Delay of “idle stop”	When genset stops, delay of “idle stop” begin after deceleration finish, and idle relay begins to work.
Delay of “ACC.”	Genset starts successfully and idle start finishes, ACC. Relay closes when ACC delay begins. “ACC failure” will alarm when the delay finish but not get the signal of ACC in a right position.
Delay of “Auxiliary Input 1”	Delay begins at the time of auxiliary input 1 closes, delay will break off when the state returns to be normal. When the input still closes after delay finishes, it will alarm.
Delay of “Auxiliary Input 2”	Delay begins at the time of auxiliary input 2 closes, delay will break off when the state returns to be normal. When the input still closes after delay finishes, it will alarm.
Delay of “Auxiliary Input 3”	Delay begins at the time of auxiliary input 3 closes, delay will break off when the state returns to be normal. When the input still closes after delay finishes,it will alarm.
Delay of “Auxiliary Input 4”	Delay begins at the time of auxiliary input 4 closes, delay will break off when the state returns to be normal. When the input still closes after delay finishes, it will alarm.
Delay of “Loss Speed”	Delay begins when no speed signal is detected during the running. If the speed signal isn't be detected until

	delay finish ,then it alarms to lose speed.
Delay of “Close time”	Relay close time is set to 0, then the output of relay will be energized constantly.
Delay of “Retransformation”	When “remote start” switch breaks and starts to delay, the controller is in a “auto” state, and it gets ready to stop after delay finishes.
Delay of Overload	Delay begins when current is exceed the alarm up limit. If the current is in normal,delay will break off. When it's still overcurrent after delaying, it will be overload to stop. When the current exceeds stop upper limit, then protect to stop without any delay.
Delay of “Overvoltage”	Delay begins when voltage exceeds the upper limit. When voltage returns to normal during the period of delay, delay will be interrupted. If the phenomena of overvoltage still exists after delay finishes, then it will be in an overvoltage protection to stop.
Delay of “Overfrequency”	Delay begins when the frequency exceed the upper limit. When voltage returns to normal during the period of delay, delay will be interrupted. If the phenomena of overfrequency still exists after delay finishes, then it will be in an overfrequency protection to stop. If the frequency is above the upper limit, then it will stop to be protected without any delay.
Delay of “Warm Up”	It provides a delay between starts and loads successfully. It can prolong the time of transferring to supply. It also can help the genset to run to supply in the best condition and avoid friction loss of genset to the minimum when it's in not an emergency situation.
Delay of “Deceleration”	Delay begins at when genset stops, and deceleration relay will close. The controller will alarm for “deceleration failure” when the right deceleration signal can not be detected after delay finishes.

5.4 Timing Start Instruction

Timing start is only valid on the automatic state of Minco430.

Autostart estimate the now date(month-day/week) to be correct or not, if it's correct, then start the machine to work or stop the machine to halt at the setting time(hour: minute).All the parameters are set to 0, then it means time

start function is not available.

Any one of month/day/week is set to 0, it means that the setting corresponds with current time. When month/day/week are all set to 0, it means timing start daily. When day and month are set to 0, it means timing start weekly. When month and week are set to 0, it means timing start monthly.

For example:

The date of timing start is set to:08-00/01 Time: 10:00-12:20. It means that the genset will starts at 10:00,and stops at 12:20 in every Monday of every week in August.

The date of timing start is set to:00-03/00 Time: 10:00-12:20. It means that the genset will starts at 10:00,and stops at 12:20 in the 3rd of every month.

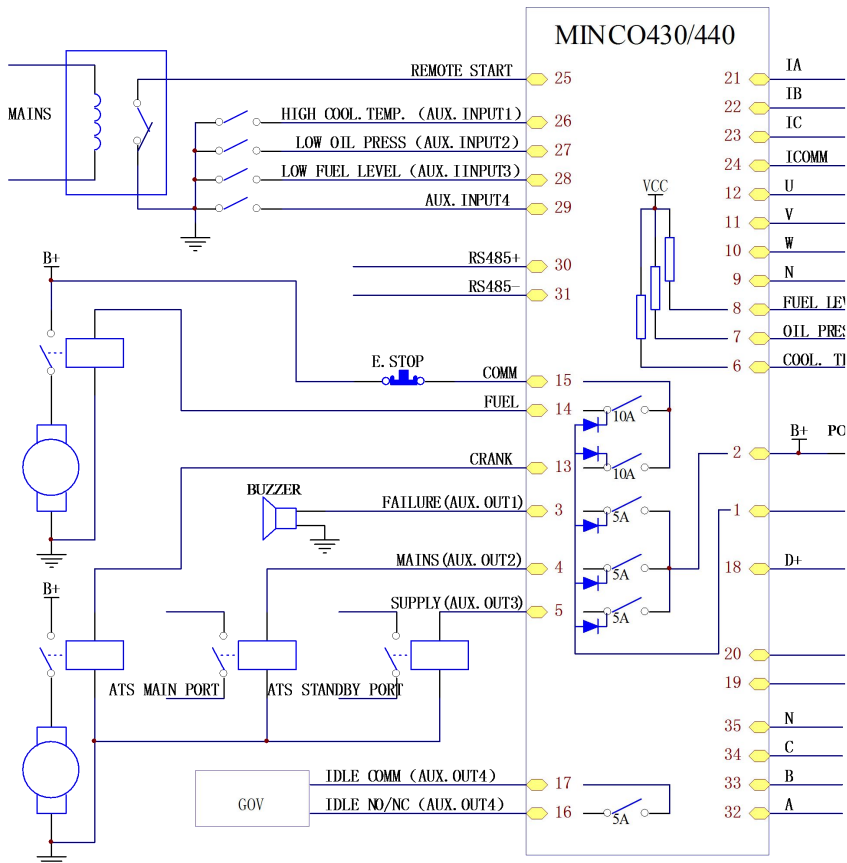
Three times interval can be set on every day, please set"0" to no-use time interval.

VI. Normal Failure and Trouble Shooting Method

Failure	Description	Solution
Manual Start Failure	 Press the  key, the green light isn't bright on the above and motor doesn't work.	Check to see whether the green light is broken or not. If the LED light isn't broken, please contact with the factory ; if the LED light is broken, please see below solution.
	 Press  key, the green light is bright on the above and the motor doesn't work.	Check "low oil pressure" in the "input port state", if it displays "0",please check whether the oil pressure sensor is ok. If display "1",the oil pressure sensor is ok. Then please  press  key, measuring the  module port 13"start"whether there's 24Vwith a multimeter, if the voltage is 24V,check whether the outside middle relay, start motor is broken, and whether

		the battery voltage is enough; If port 13 no output, the module might be damaged.
Auto Start Failure	 Module is in state , inspection "remote start" have input, the "remote start" state light is bright on and the motor doesn't work.	Check the menu of "remote start" in the "input state", if the "remote start" display "0"means that the outside timer etc module relay is broken cause didn't receive the input signal; If display "1",the module might be broken.
	 Module in state , inspection "remote start" have input, the "remote start" state light is bright on and the motor doesn't work;	Check the oil pressure sensor; switch to the manual start, check whether there are output signal of the port 13-"start",the outside components and the battery voltage.
Wheel tooth is fighting when starts	Start successfully but motor keep running, the wheel tooth is fighting.	Lower down the trip speed;
Onload current displays incorrect	Current ratio in system parameter setting is wrong.	Reset the current ratio.

VII. Outside Wiring diagram





VIII. Controller Front Panel Diagram:

Minco 430 and Minco440

